VTPEH 6270 - HW 3

Cameron Lincoln - cal386

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0.1 From the original instructions...

`*This section is copy/pasted directly from the instructions*`

This document is for instructions only. Please create your own R markdown document as a basis for the rest of the project. The document should respect the following criteria:

- Is a PDF.
- Include headers as relevant, and a table of content.
- Include all R code.
- Include all figures.
- Console output may or may not be printed, as relevant.
- Is understandable and pleasant to navigate (e.g., avoid printing full data set within the report). – **Please note that formatting errors will be penalized**

Sections from previous reports can be re-used if relevant. Any section of the report can be revised as the project develops. The goal of this Check Point is to further data set exploration, and gain experience production publication-quality reports with R markdown. It is also an opportunity to prepare for future reports focused on statistical analyses.

0.1.1 Research Question and Background

`*This section is copy/pasted directly from the instructions*`

- **Research Question**. Select one focused research question that examines the association between 2+ variables A and B (e.g., "Does X influence Y?", "Is Y associated with Z?", "Does Y occur more frequently in group 1 than group 2?").
- **Scientific Plausibility and Context**. Explain why this association makes sense from a scientific perspective. Include at least 3 relevant citations that provide background for your research question and expected outcome if relevant. Use proper citation formatting (e.g., see https://yihui.org/rmarkdown-cookbook/bibliography).

0.1.2 Data Preparation

For the following sections, we require that at least one of the considered variables (B) to be categorical for summary and visualization purpose. If the research question focuses on continuous variables, one of the variables can be discretized (see below).

If relevant, include any required data cleaning, subsetting or transformation here. Data transformation can include:

- **Derived Variables**. Create meaningful derived variables relevant to your research question (e.g., BMI from height/weight, time intervals from dates, ratios, z-scores, log transformations)
- **Categorization and Discretization**. If analyzing continuous variables, consider creating meaningful categories or classes for summary and visualization purposes.

0.1.3 Grouped Summary Statistics

- **Stratified Analysis**. Create comprehensive tables of **descriptive statistics** for your variable A, stratified by your categorical variable B.
- **Interpretation**. Discuss what these statistics suggest about potential associations.

0.1.4 Comparative Visualizations

- **Association Visualization**. Create plots that directly address your research question using appropriate visualization types to explore the potential association between A and B.
- **Visuals for Publication**. Save one visual as a PDF following the *Nature* formatting guidelines.
- **Interpretation**. For each visualization, provide scientific interpretation of what the patterns suggest about your research question

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1 Introduction

1.1 Research Question

This will be a continuation of my earlier exploration of my lead in NYS data set from HW2. My original question was:

Are there specific areas/regions of NYS that are likely to have more rates of childhood blood lead levels? If so, where are they clustered? How large is the difference in rates?

With the new requirements of this assignment, I think I can explore the following variables:

- Are there certain **counties** that have less/more **reported cases of elevated blood lead level** among children?
- Are there changes that we can see over time **(years)** where the **number of reported cases** increased?

1.2 Scientific Plausibility and Context

As alluded to in my last homework, I am slightly interested in the link between geography and blood lead levels in children. Broadly, there is a public campaign to ensure that children are screened for blood lead levels routinely. Due to a variety of social, environmental, and economic factors, children of some backgrounds are more at risk of exposure to lead, which can have devastating long term effects on development and overall health.[1] States typically have standards on providing blood lead testing for children. In the state of New York, all children are supposed to undergo regular checkups to screen for elevated blood lead testing.[2] These guidelines change frequently, as recently as 2019 for medical providers, and with rental registry laws in 2025.[2] Around the country, there have been national efforts to improve drinking water via the 2021 Infrastructure Investment and Jobs Act, these projects are up to the individual states to determine what projects get priority first.[3] For New York State, municipalities with high populations of people of color and lower incomes are likely to be prioritized by the lead service line replacement program, as these individuals may be more at risk for blood lead levels.[3]

References

1. CDC. CDC Updates Blood Lead Reference Value. Childhood Lead Poisoning Prevention. Published April 2, 2024. https://www.cdc.gov/lead-prevention/php/news-features/updates-blood-lead-reference-value.html
2. New York State Department of Health. Childhood Lead Poisoning Prevention. ny.gov. Accessed February 19, 2026. https://www.health.ny.gov/environmental/lead/
3. Lead Service Lines in NYC Disproportionately Impact Hispanic/Latino Communities and Children Already At Risk. Columbia University Mailman School of Public Health. Published August 30, 2023. https://www.publichealth.columbia.edu/news/lead-service-lines-nyc-disproportionately-impact-hispanic-latino-communities-children-already-risk-lead-exposure

2 Data Preparation

This section was adapted from my last homework submission

The data set I will be working with is from by the New York State Department of Health. Per state regulation, doctors are required to do blood level testing for all children at specific time intervals. This data

includes all children tested for lead from 2000 to March 6, 2024. Important to note, this data does not include information from NYC.

There are nullification qualifications to protect patient anonymity in cases where positive result/number of tests per Zip code is low. Tests are pulled by year, and are reported on the Zip code, geographic center, and county level.

To see the more information about the data set, please visit the website provided: https://health.data.ny.gov/Health/Childhood-Blood-Lead-Testing-and-Elevated-Incidenc/d54z-enu8/about_data

2.1 Derived Variables

```
# Set up the work directory,

setwd("/Users/clincoln/Desktop/vtpeh-6270")
getwd()

# then load some packages,
library(ggplot2)
library(dplyr)
library(tidyr)
library(knitr)
library(kableExtra)
library(gtsummary)

# store the data, and then ...
data_lead <-
  read.csv(
    "Childhood_Blood_Lead_Testing_and_Elevated_Incidence_by_Zip_Code__Beginning_2000_20260205.csv")

# and then preview the data.
head(data_lead)
colnames(data_lead)
summary(data_lead)

# Some of these data are classed as integers, when they should be categories.
data_lead$County.Code <- as.character(data_lead$County.Code)
data_lead$Zip <- as.character(data_lead$Zip)

# And vice-versa
data_lead$Tests <- as.numeric(data_lead$Tests)
data_lead$Less.than.5.mcg.dL <- as.numeric(data_lead$Less.than.5.mcg.dL)

# So from the data set, I want to use: county, county code, year, ZIP code,
# number of tests performed, percentage, and rate per 1,000
# Before I go in and change the names, I'm going back to the data source to
# double check that my re-names are in alignment with the codebook.

# Renaming all my columns to be consistent. NOTE: Percent needs to be
# "New_Case_Percentage," since these ONLY count newly diagnosed children who
# had a positive blood Pb level.

data_lead_tidy <- data_lead
```

```r
data_lead_tidy <- data_lead %>%
  select(-"Less.than.5.mcg.dL", -"X5.10.mcg.dL",
         -"X10...15.mcg.dL", -"X15...mcg.dL")

data_lead_tidy <- data_lead_tidy %>%
  rename("County_Code" = "County.Code",
         "Total_Elevated_Blood_Levels" = "Total.Elevated.Blood.Levels.",
         "New_Case_Percentage" = "Percent",
         "Rate_per_1000" = "Rate.per.1.000",
         "ZIP_Location" = "Zip.Code.Location",
         "County_Location" = "County.Location")
# I'm going to exclude the 0-data
table(data_lead_tidy$Rate_per_1000)

non0rate_data <- data_lead_tidy[
  data_lead_tidy$Rate_per_1000 > 0,]

table(non0rate_data$Rate_per_1000)

# This should give us a much cleaner data set.
head(non0rate_data)
summary(non0rate_data)
```

2.2 Categorization and Discretization

```r
# I have to make a new table with the data variables, types, and classes.

data_lead_vars <- data.frame(
  Variable = c("County",
               "County_Code",
               "Year",
               "Zip",
               "Tests",
               "Total_Elevated_Blood_Levels",
               "New_Case_Percentage",
               "Rate_per_1000",
               "ZIP_Location",
               "County_Location"),
  Type = c("Categorical",
           "Categorical",
           "Discrete",
           "Categorical",
           "Discrete",
           "Continuous",
           "Continuous",
           "Continuous",
           "Categorical",
           "Categorical"),
  Class = c("Character",
            "Character",
            "Integer",
```

Table 1: Table with all the variables by their class and type

Variable	Type	Class
County	Categorical	Character
County_Code	Categorical	Character
Year	Discrete	Integer
Zip	Categorical	Character
Tests	Discrete	Numeric
Total_Elevated_Blood_Levels	Continuous	Integer
New_Case_Percentage	Continuous	Numeric
Rate_per_1000	Continuous	Integer
ZIP_Location	Categorical	Character
County_Location	Categorical	Character

```
            "Character",
            "Numeric",
            "Integer",
            "Numeric",
            "Integer",
            "Character",
            "Character")
)

kbl(data_lead_vars[1:10,], booktabs = T,
    caption = "Table with all the variables by their class and type") %>%
  kable_styling(latex_options = c("striped", "scale_down"))
```

Here, we see that we have a a few categorical *AND* continuous variables. To answer the questions set out in the introduction, I will need to pick:

- **County**, already a categorical variable with 57 unique counties.
- **Total_Elevated_Blood_Levels,** a discrete variable with the number of individuals who tested positive for elevated blood levels.
  - NOTE: In this homework submission, I do want to include 0-values, as there were years with high testing with no reported cases. This might be good to analyze.

3 Group Summary Statistics

3.1 Stratified Analysis

Create comprehensive tables of **descriptive statistics** for your variable A, stratified by your categorical variable B.

```
# I need to create tables of descriptive statistics
# Per the instruction, I'll assume that variable A needs to be discritized, and
# variable B will need to be categorical.

# varA = Total_Elevated_Blood_Levels, and varB = County
```

```r
# should report the median and IQRs given that some of the county
# data is wonky, not normally distrobuted at all.

non0rate_data %>%
  group_by(County) %>%
  summarise(
    median = median(Total_Elevated_Blood_Levels, na.rm = TRUE),
    q1 = quantile(Total_Elevated_Blood_Levels, 0.25, na.rm = TRUE),
    q3 = quantile(Total_Elevated_Blood_Levels, 0.75, na.rm = TRUE),
    mean = mean(Total_Elevated_Blood_Levels, na.rm = TRUE),
    sd = sd(Total_Elevated_Blood_Levels, na.rm = TRUE),
    .groups = "drop"
  )
```

```r
# The following code-string was generated by ChatGPT

library(dplyr)
library(gt)

non0rate_data %>%
  group_by(County) %>%
  summarise(
    median = median(Total_Elevated_Blood_Levels, na.rm = TRUE),
    q1 = quantile(Total_Elevated_Blood_Levels, 0.25, na.rm = TRUE),
    q3 = quantile(Total_Elevated_Blood_Levels, 0.75, na.rm = TRUE),
    mean = mean(Total_Elevated_Blood_Levels, na.rm = TRUE),
    sd = sd(Total_Elevated_Blood_Levels, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  mutate(
    `Median (Q1, Q3)` = sprintf(
      "%.0f (%.0f, %.0f)",
      round(median),
      round(q1),
      round(q3)
    ),
    `Mean (SD)` = sprintf(
      "%.0f (%.0f)",
      round(mean),
      round(sd)
    )
  ) %>%
  select(County, `Median (Q1, Q3)`, `Mean (SD)`) %>%
   gt() %>%
  tab_header(
    title = md("**Number of Elevated Pediatric Blood Lead Tests
               Reported in New York State (Excluding New York City),
               2000-2020**"),
      subtitle = "County-level summary statistics"

  )
```

3.2 Interpretation

The table generated (attached on the last page) shows the mean and median amount of tests performed each year that indicate elevated lead in children. I chose to report the median with IQRs since many counties distribution of positive cases did not seem normally distributed. With this data, it looks like you can see that certain areas have higher medians of positive elevated pediatric blood lead results. For instance, Chautauqua, Dutches, and Oneida Counties are some counties with higher medians than others. It should be noted that Oneida has a wide IQR, as well as a higher mean.

It would be interesting to investigate further if this is truly because Oneida County has higher rates of lead exposure for children, OR if this is an artifact of the data. We are lumping together 20 years of elevated pediatric blood lead testing cases. It may be that some counties initially had higher blood lead levels, then changes in reporting/exposure could've changed this value. We could see if that is the case by comparing this value over third variable: Year.

I noticed that some counties reported multiple values per year by geographical subarea (ZIP code), where as others only reported one testing value for the entire county. This could be another potential shortcomming in comparing data values.

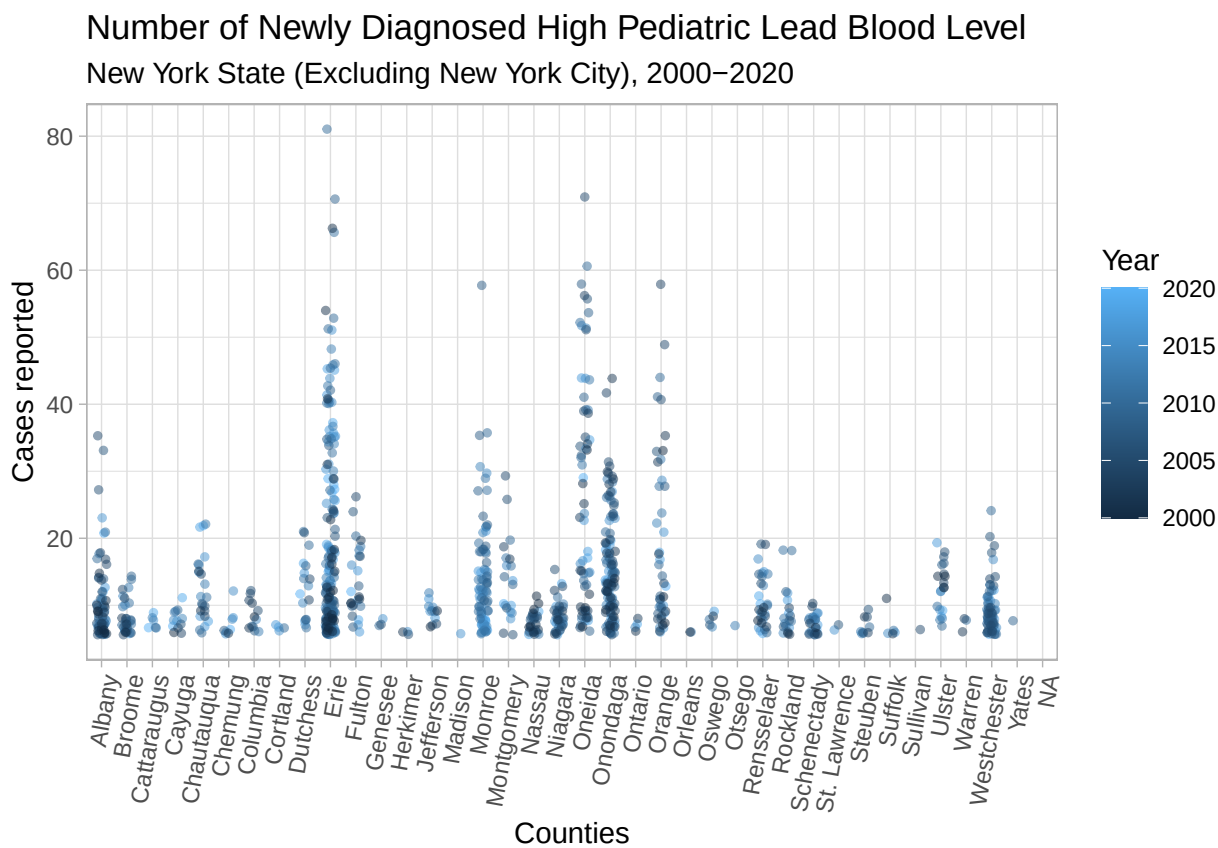
4 Comparative Visualizations

4.1 Association Visualizations

In comparing these values, we could do a two-variable plot, where we're looking at county versus number of elevated blood level tests. Then, modify it to show the third variable: time of reporting (year).

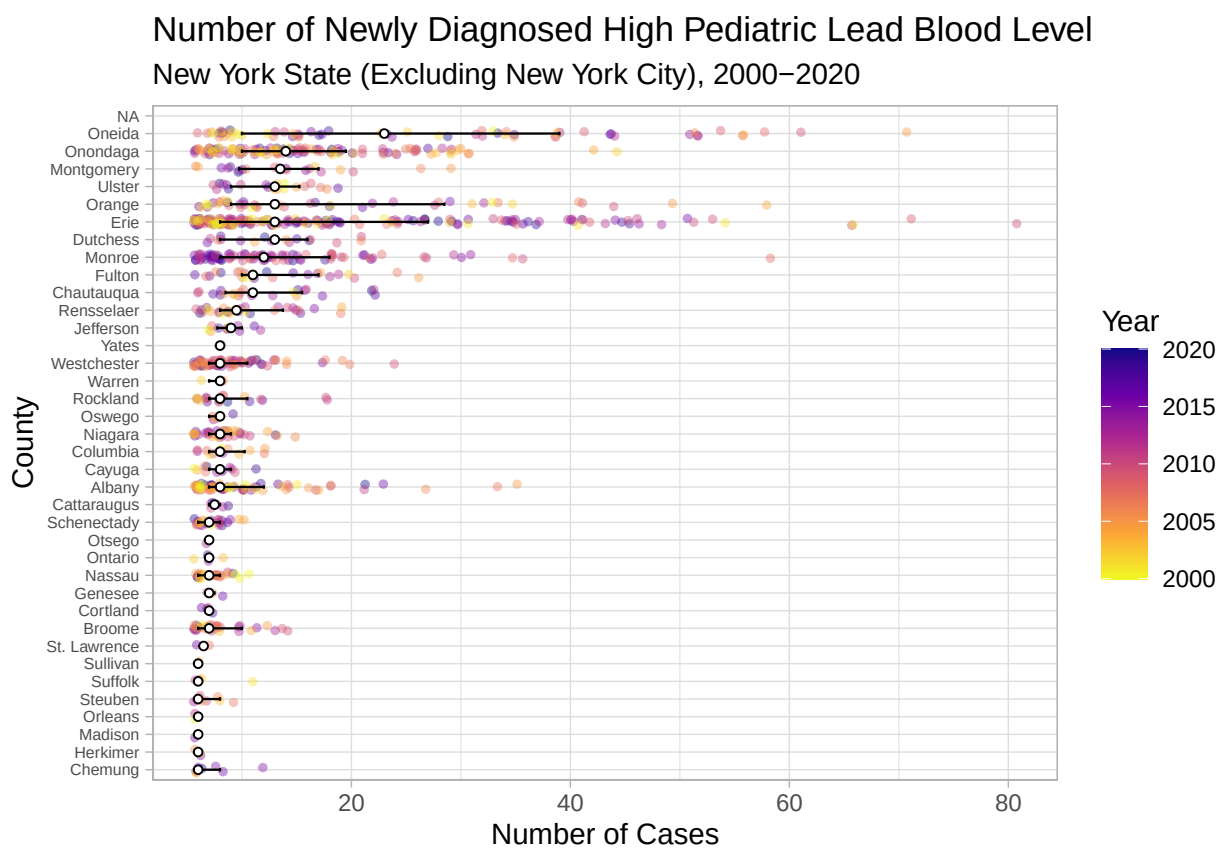
4.2 Visuals for Publication

```
# This was my attempt at trying to do this graph.
ggplot(non0rate_data, mapping = aes(x = County,
                                    y = Total_Elevated_Blood_Levels,
                                    color = Year))+
  geom_jitter(alpha = 0.5,
              size= 1,
              width = 0.2)+
  scale_y_continuous()+
  labs(title =
         "Number of Newly Diagnosed High Pediatric Lead Blood Level",
       subtitle ="New York State (Excluding New York City), 2000-2020",
       x = "Counties",
       y = "Cases reported")+
  theme_light()+
  theme(axis.text.x = element_text(angle = 80, hjust = 1))
```



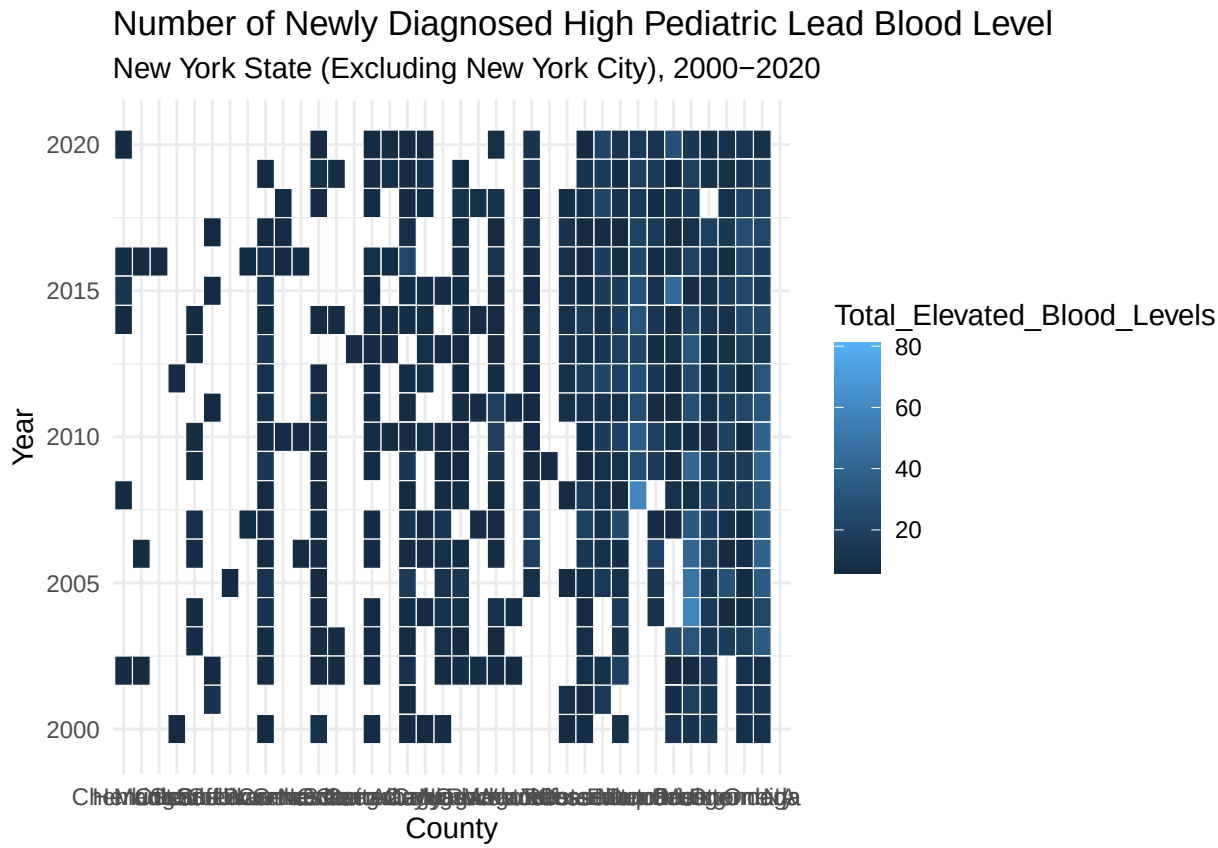

```r
# This is what ChatGPT generated with edits done by me
county_order <- non0rate_data %>%
  group_by(County) %>%
  summarise(median_val = median(Total_Elevated_Blood_Levels, na.rm = TRUE)) %>%
  arrange(median_val) %>%
  pull(County)

non0rate_data$County <- factor(non0rate_data$County, levels = county_order)

county_summary <- non0rate_data %>%
  group_by(County) %>%
  summarise(
    median_val = median(Total_Elevated_Blood_Levels, na.rm = TRUE),
    q1 = quantile(Total_Elevated_Blood_Levels, 0.25, na.rm = TRUE),
    q3 = quantile(Total_Elevated_Blood_Levels, 0.75, na.rm = TRUE),
    .groups = "drop"
  )

ggplot(non0rate_data, aes(x = County, y = Total_Elevated_Blood_Levels, color = Year)) +
  geom_jitter(alpha = 0.4, size = 1, width = 0.2) +
  geom_errorbar(
    data = county_summary,
    aes(x = County, ymin = q1, ymax = q3),
    inherit.aes = FALSE,
    color = "black",
    width = 0.2,
```

```r
    size = 0.4
  ) +
  geom_point(
    data = county_summary,
    aes(x = County, y = median_val),
    inherit.aes = FALSE,
    shape = 21,
    fill = "white",
    color = "black",
    size = 1.2
  ) +
  scale_color_viridis_c(option = "plasma", direction = -1, name = "Year") +
  coord_flip() +
  labs(
    title =
      "Number of Newly Diagnosed High Pediatric Lead Blood Level",
    subtitle = "New York State (Excluding New York City), 2000-2020",
    x = "County",
    y = "Number of Cases"
  ) +
  theme_light() +
  theme(axis.text.y = element_text(size = 6))
```



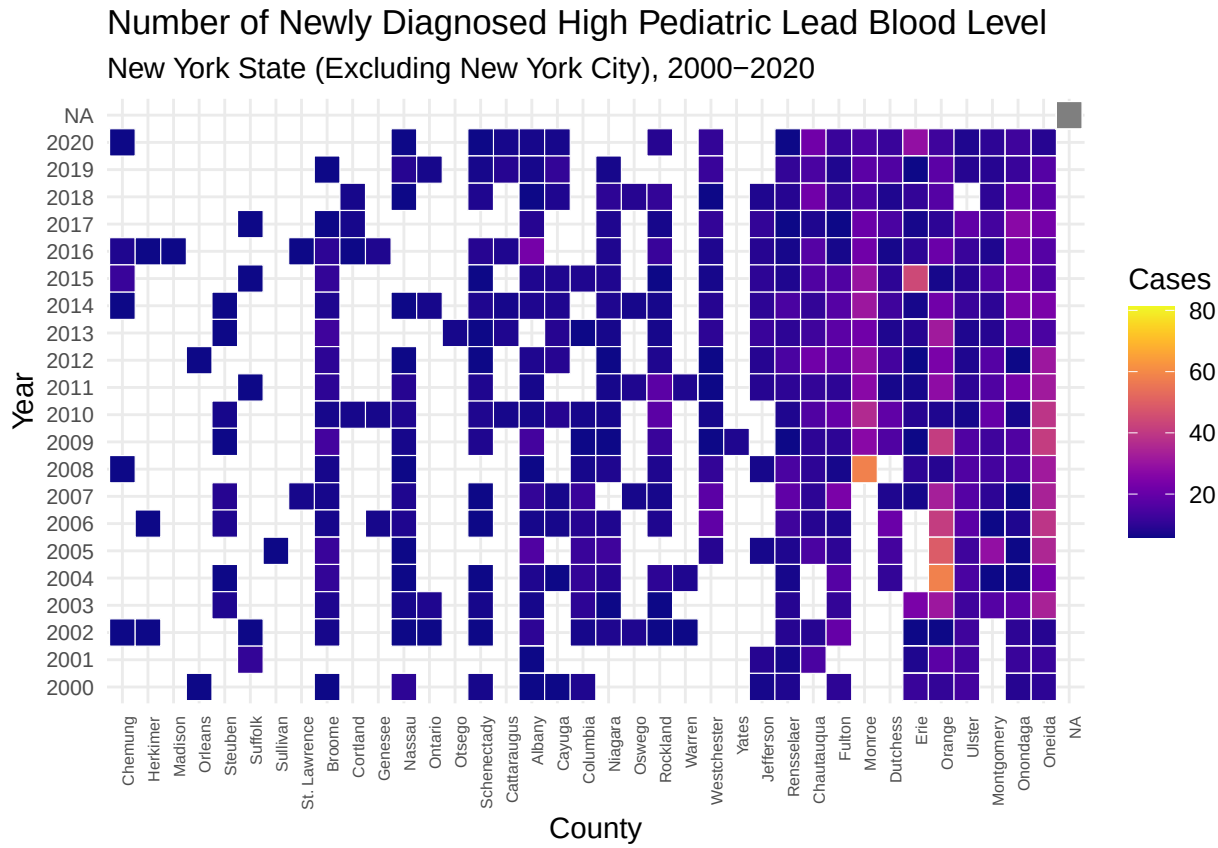

```r
# My attempt at a heatmap
ggplot(non0rate_data, aes(x = County,
```

```
                          y = Year,
                          fill = Total_Elevated_Blood_Levels)) +
  geom_tile(color = "white") +
  labs(
    title = "Number of Newly Diagnosed High Pediatric Lead Blood Level",
    subtitle = "New York State (Excluding New York City), 2000-2020",
    x = "County",
    y = "Year"
  ) +
  theme_minimal()
```




```
# This is what ChatGPT generated with edits done by me
library(viridis)
ggplot(non0rate_data, aes(x = County, y = factor(Year), fill = Total_Elevated_Blood_Levels)) +
  geom_tile(color = "white") +
  scale_fill_viridis(option = "plasma", name = "Cases") +
  labs(
    title = "Number of Newly Diagnosed High Pediatric Lead Blood Level",
    subtitle = "New York State (Excluding New York City), 2000-2020",
    x = "County",
    y = "Year"
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_text(angle = 90, hjust = 1, size = 6),
```

```
    axis.text.y = element_text(size = 8)
  )
```



4.3 Interpretation

This jitter plot with median and IQR allows us to see both the amount of positive testing done by the county, as well as an associated colored time with each reported dot. Immediate, you can see that there is a difference in reporting times between counties. For example, Monroe County's plots are very light, indicating that they've only started to report elevated pediatric blood level tests in the 2010s. Similarly, you can see with Oneida, there's a wide range of different reported cases per year. With the IQRs and medians visible on this plot, we can see how some of our summary statistics might not tell the full picture of the data.

Comparative, you can look at this data as a heatmap. Note that several counties have sparse data, which could be indicative of (a) low amounts of pediatric exposure to lead, (b) poor data quality, screening and/or reporting issues, or (c) both. We have other data in `data_lead_tidy$Tests` that lets us know how many tests are performed by the county. However, because we are only reporting number of newly diagnosed cases of pediatric lead exposure, we assume that all blank tiles on this heat map mean 0 new cases of lead exposure. From here, we can also see that Oneida County has had higher levels of pediatric blood lead levels from 2003 to 2010s. We can also confirm our earlier thought that Monroe County only had started to report cases starting from the 2010s. On this heat map, we can see that Monroe initially started with a higher number of new cases in 2008., and then has been consistently reporting a non-zero amount of cases ever since.

5 AI Disclosure

All of the text written to explain my thinking is 100% human-generated. Given some personal events, I needed to request an extension. Typically, I would reach out for help from friends or peers who are more fluent in R than me. I am not confident in my debugging skills (I'm still re-learning how to use R after several years), and I used ChatGPT to help fix 3 coding errors. Included are my honest attempts at trying to display the data, as well as the altered AI-generated version with minor edits I made. I do want to learn these skills, as using AI to code (imho) doesn't make me feel I can reproduce this code organically.

Number of Elevated Pediatric Blood Lead Tests Reported in New York State (Excluding New York City), 2000–2020

County-level summary statistics

County	Median (Q1, Q3)	Mean (SD)
Albany	8 (7, 12)	11 (6)
Broome	7 (6, 10)	8 (2)
Cattaraugus	8 (7, 8)	8 (1)
Cayuga	8 (7, 9)	8 (1)
Chautauqua	11 (8, 16)	12 (5)
Chemung	6 (6, 8)	7 (2)
Columbia	8 (7, 10)	9 (2)
Cortland	7 (7, 7)	7 (0)
Dutchess	13 (8, 16)	13 (5)
Erie	13 (8, 27)	19 (15)
Fulton	11 (10, 17)	13 (5)
Genesee	7 (7, 8)	7 (1)
Herkimer	6 (6, 6)	6 (0)
Jefferson	9 (8, 10)	9 (2)
Madison	6 (6, 6)	6 (NA)
Monroe	12 (8, 18)	14 (9)
Montgomery	14 (10, 17)	14 (6)
Nassau	7 (6, 8)	7 (1)
Niagara	8 (7, 9)	8 (2)
Oneida	23 (10, 39)	27 (17)
Onondaga	14 (10, 20)	16 (8)
Ontario	7 (7, 7)	7 (1)
Orange	13 (9, 28)	19 (13)
Orleans	6 (6, 6)	6 (0)
Oswego	8 (7, 8)	8 (1)
Otsego	7 (7, 7)	7 (NA)
Rensselaer	10 (8, 14)	11 (4)
Rockland	8 (7, 10)	9 (4)
Schenectady	7 (6, 8)	7 (1)
St. Lawrence	6 (6, 7)	6 (1)
Steuben	6 (6, 8)	7 (1)
Suffolk	6 (6, 6)	7 (2)
Sullivan	6 (6, 6)	6 (NA)
Ulster	13 (9, 15)	13 (4)
Warren	8 (7, 8)	7 (1)
Westchester	8 (7, 10)	9 (4)
Yates	8 (8, 8)	8 (NA)
NA	NA (NA, NA)	NaN (NA)